ASSIGNMENT NO.4

Question 1: Print Numbers from 1 to N

Problem Statement:

Write a Java program that asks the user for a number N and then prints the numbers from 1 to N using a for loop.

Code:

```
import java.util.Scanner;
public class PrintNumbers {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int n = sc.nextInt();

        for (int i = 1; i <= n; i++) {
            System.out.print(i+"  ");
        }
    }
}
```

Expected Output:

Enter a number: 10

1 2 3 4 5 6 7 8 9 10

Question 2: Print Multiples of 3 between 1 and N

Problem Statement:

Write a Java program that asks the user for a number N and prints all the multiples of 3 between 1 and N using a for loop.

Code:

```
import java.util.Scanner;
public class MultiplesOfThree {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int n = sc.nextInt();

        for (int i = 1; i <= n; i++) {
            if (i % 3 == 0) {
                System.out.print(i+"  ");
            }
```

```
        }
    }
}
```

Expected Output:
Enter a number: 20
3 6 9 12 15 18

Question 3: Calculate the Factorial of a Number
Problem Statement:
Write a Java program that asks the user for a number N and calculates the factorial of N using a for loop.

Code:
```
import java.util.Scanner;
public class Factorial {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int n = sc.nextInt();

        long factorial = 1;
        for (int i = 1; i <= n; i++) {
            factorial *= i;
        }
        System.out.println("Factorial of " + n + " is " + factorial);
    }
}
```

Expected Output:
Enter a number: 5
Factorial of 5 is 120

Question 4: Print Even Numbers from 1 to N
Problem Statement:
Write a Java program that asks the user for a number N and prints all the even numbers from 1 to N using a for loop.

Code:

```
import java.util.Scanner;
public class EvenNumbers {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int n = sc.nextInt();

        for (int i = 1; i <= n; i++) {
            if (i % 2 == 0) {
                System.out.print(i+" ");
            }
        }
    }
}
```

Expected Output:

```
Enter a number: 10
2 4 6 8 10
```

Question 5: Sum of Odd Numbers between 1 and N

Problem Statement:

Write a Java program that asks the user for a number N and calculates the sum of all odd numbers between 1 and N using a for loop.

Code:

```
import java.util.Scanner;
public class OddSum {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int n = sc.nextInt();

        int sum = 0;
        for (int i = 1; i <= n; i++) {
            if (i % 2 != 0) {
                sum += i;
            }
        }
        System.out.println("The sum of odd numbers from 1 to " + n + " is: " + sum);
    }
}
```

Expected Output:
Enter a number: 10
The sum of odd numbers from 1 to 10 is: 25

Question 6: Print All Elements of an Array

Problem Statement:
Write a Java program that uses a for-each loop to print all elements of an integer array. The program should ask the user to input 5 integers, store them in an array, and then print all the elements using a for-each loop.

Code:
```
import java.util.Scanner;
public class PrintArrayElements {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);
      int[] numbers = new int[5];

      System.out.print("Enter 5 integers: ");
      for (int i = 0; i < numbers.length; i++) {
         numbers[i] = sc.nextInt();
      }

      for (int num : numbers) {
         System.out.println(num+" ");
      }
   }
}
```

Expected Output:
Enter 5 integers: 3 7 12 5 8
3 7 12 5 8

Question 7: Find the Sum of All Elements in an Array

Problem Statement:
Write a Java program that uses a for-each loop to calculate the sum of all elements in a given integer array. The program should ask the user to input 5 integers, store them in an array, and then compute the sum of these numbers using the for-each loop.

Code:

```
import java.util.Scanner;
public class SumOfArrayElements  {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int[] numbers = new int[5];

        System.out.print("Enter 5 integers: ");
        for (int i = 0; i < numbers.length; i++) {
            numbers[i] = sc.nextInt();
        }
        int sum = 0;
        for (int num : numbers) {
            sum += num;
        }
        System.out.println("The sum of all numbers is: " + sum);
    }
}
```

Expected Output:

Enter 5 integers: 4 6 8 2 10
The sum of all numbers is: 30

Question 8: Print All Names in a String Array

Problem Statement:

Write a Java program that uses a for-each loop to print all the names stored in a String array. The program should ask the user to input 4 names, store them in an array, and then print each name using the for-each loop.

Code:

```
import java.util.Scanner;
public class PrintNames {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        String[] names = new String[4];
        System.out.print("Enter 4 names: ");
        for (int i = 0; i < names.length; i++) {
            names[i] = sc.next();
        }
        for (String name : names) {
            System.out.println(name);
        }
    }
}
```

Expected Output:
Enter 4 names: Major Aditya Nihal Shivam
Major
Aditya
Nihal
Shivam

Question 9: Find the Largest Element in an Array

Problem Statement:
Write a Java program that asks the user to input 5 integers, stores them in an array, and then finds and prints the largest element in the array. (Explore in-built method to solve this)

Code:
```
import java.util.Scanner;
import java.util.Arrays;

public class LargestElement {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int[] numbers = new int[5];

        System.out.print("Enter 5 integers: ");
        for (int i = 0; i < numbers.length; i++) {
            numbers[i] = sc.nextInt();
        }
        int largest = Arrays.stream(numbers).max().getAsInt();
        System.out.println("The largest element is: " + largest);
    }
}
```

Expected Output:
Enter 5 integers: 12 45 67 23 89
The largest element is: 89

Question 10: Find the Average of Elements in an Array

Problem Statement:
Write a Java program that asks the user to input 5 integers, stores them in an array, and then calculates and prints the average of the elements in the array.

Code:
```
import java.util.Scanner;
public class AverageOfArray {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int[] numbers = new int[5];

        System.out.print("Enter 5 integers: ");
        for (int i = 0; i < numbers.length; i++) {
            numbers[i] = sc.nextInt();
        }

        int sum = 0;
        for (int num : numbers) {
            sum += num;
        }
        double average = (double) sum / numbers.length;
        System.out.println("The average of the numbers is: " + average);
    }
}
```

Expected Output:
```
Enter 5 integers: 10 20 30 40 50
The average of the numbers is: 30.0
```

Question 11: Count Positive and Negative Numbers in an Array

Problem Statement:

Write a Java program that asks the user to input 6 integers, stores them in an array, and then counts how many positive and negative numbers are present in the array.

Code:
```
import java.util.Scanner;

public class CountPositiveNegative {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int[] numbers = new int[6];

        System.out.print("Enter 6 integers: ");
        for (int i = 0; i < numbers.length; i++) {
            numbers[i] = sc.nextInt();
        }
```

```
        int positiveCount = 0;
        int negativeCount = 0;

        for (int num : numbers) {
            if (num > 0) {
                positiveCount++;
            } else if (num < 0) {
                negativeCount++;
            }
        }
        System.out.println("Positive numbers: " + positiveCount);
        System.out.println("Negative numbers: " + negativeCount);
    }
}
```

Output:
Enter 6 integers: -5 3 7 -2 0 8
Positive numbers: 3
Negative numbers: 2

Question 12: Sort an Array in Ascending Order
Problem Statement:
Write a Java program that asks the user to input 5 integers, stores them in an array, and then sorts the array in ascending order using the Arrays.sort() method. After sorting, print the sorted array.

Code:
```
import java.util.Scanner;
import java.util.Arrays;
public class SortArray {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int[] numbers = new int[5];
        System.out.print("Enter 5 integers: ");
        for (int i = 0; i < numbers.length; i++) {
            numbers[i] = sc.nextInt();
        }
        Arrays.sort(numbers);
        System.out.print("Sorted array: ");
        for (int num : numbers) {
            System.out.print(num+" ");
        }
    }
}
```

Expected Output:
Enter 5 integers: 12 45 23 8 90
Sorted array: 8 12 23 45 90

Question 13: Check if an Array Contains a Specific Element

Problem Statement:
Write a Java program that asks the user to input 5 integers, stores them in an array, and then checks whether a specific number (input by the user) is present in the array using the Arrays.asList() method. If the number is found, print "Found", otherwise print "Not Found".

Code:
```
import java.util.Scanner;
import java.util.Arrays;

public class ContainsElement  {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        Integer[] numbers = new Integer[5];

        System.out.print("Enter 5 integers: ");
        for (int i = 0; i < numbers.length; i++) {
            numbers[i] = sc.nextInt();
        }
        System.out.print("Enter the number to search: ");
        int target = sc.nextInt();

        boolean found = Arrays.asList(numbers).contains(target);

        System.out.println(found ? "Found" : "Not Found");
    }
}
```
Expected Output:
Enter 5 integers: 10 20 30 40 50
Enter the number to search: 30
Found

Question 14: Find the Index of an Element in an Array

Problem Statement:
Write a Java program that asks the user to input 5 integers, stores them in an array, and then finds the index of a specific number (input by the user) using the Arrays.binarySearch() method. If the number is found, print the index, otherwise print "Not Found".

Code:

```
import java.util.Scanner;
import java.util.Arrays;
public class FindIndexBinarySearch  {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int[] numbers = new int[5];

        System.out.print("Enter 5 integers: ");
        for (int i = 0; i < numbers.length; i++) {
            numbers[i] = sc.nextInt();
        }
        System.out.print("Enter the number to search: ");
        int target = sc.nextInt();

        int index = Arrays.binarySearch(numbers, target);

        if (index >= 0) {
            System.out.println("The number " + target + " is found at index " + index);
        } else {
            System.out.println("Not Found");
        }
    }
}
```

Expected Output:

Enter 5 integers: 5 10 15 20 25
Enter the number to search: 15
The number 15 is found at index 2

Question 15: Write a program to print the following pattern:

```
1
2*2
3*3*3
4*4*4*4
5*5*5*5*5
5*5*5*5*5
4*4*4*4
3*3*3
2*2
```

Code:

```
public class Pattern1 {
    public static void main(String[] args) {
        int n = 5;

        for (int i = 1; i <= n; i++) {
            for (int j = 1; j <= i; j++) {
                System.out.print(i);
                if (j < i) {
                    System.out.print("*");
                }
            }
            System.out.println();
        }

        for (int i = n; i >= 1; i--) {
            for (int j = 1; j <= i; j++) {
                System.out.print(i);
                if (j < i) {
                    System.out.print("*");
                }
            }
            System.out.println();
        }
    }
}
```

Output:

```
1
2*2
3*3*3
4*4*4*4
5*5*5*5*5
5*5*5*5*5
4*4*4*4
3*3*3
2*2
1
```

Question 16: Write a program to print the following pattern:

1
1*2
1*2*3
1*2*3*4
1*2*3*4*5

Code:

```
public class Pattern2{
    public static void main(String[] args) {
        int n = 5;

        for (int i = 1; i <= n; i++) {
            for (int j = 1; j <= i; j++) {
                System.out.print(j);
                if (j < i) {
                    System.out.print("*");
                }
            }
            System.out.println();
        }
    }
}
```

Output:

1
1*2
1*2*3
1*2*3*4
1*2*3*4*5

Question 17: Write a program to print the following pattern:

1
1*3
1*3*5
1*3*5*7
1*3*5*7*9

Code:

```
public class Pattern3{
    public static void main(String[] args) {
        int n = 5;

        for (int i = 1; i <= n; i++) {
            for (int j = 1; j <= i; j++) {
                int value = 2 * j - 1;
                System.out.print(value);
```

```
            if (j < i) {
                System.out.print("*");
            }
        }
        System.out.println();
    }
  }
}
```

Output:

1
1*3
1*3*5
1*3*5*7
1*3*5*7*9

Question 18: Write a program to print the following pattern:

11111
22222
33333
44444
55555

Code:

```
public class Pattern4{
  public static void main(String[] args) {
    int n = 5;
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= n; j++) {
            System.out.print(i);
        }
        System.out.println();
    }
  }
}
```

Output:

11111
22222
33333
44444
55555

Question 19: Write a program to print the following pattern:

1
22
333
4444
55555

Code:

```
public class Pattern5{
    public static void main(String[] args) {
        int n = 5;
        for (int i = 1; i <= n; i++) {
            for (int j = 1; j <= i; j++) {
                System.out.print(i);
            }
            System.out.println();
        }
    }
}
```

Output:

1
22
333
4444
55555

Question 20: Write a program to print the following pattern:

1
12
123
1234
12345

Code:

```
public class Pattern6{
    public static void main(String[] args) {
        int n = 5;
        for (int i = 1; i <= n; i++) {
            for (int j = 1; j <= i; j++) {
                System.out.print(j);
            }
            System.out.println();
        }
    }
}
```

Output:

```
1
12
123
1234
12345
```

Question 21: Write a program to print the following pattern:

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

Code:

```
public class Pattern7{
    public static void main(String[] args) {
        int n = 5;
        int num = 1;
        for (int i = 1; i <= n; i++) {
            for (int j = 1; j <= i; j++) {
                System.out.print(num+" ");
               num+=1;
            }
            System.out.println();
        }
    }
}
```

Output:

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

Question 22: Write a program to print the following pattern:

```
******
*    *
*    *
*    *
*    *
******
```

Code:

```
public class Pattern8{
    public static void main(String[] args) {
        int n = 6;
        for (int i = 0; i < n; i++) {
            for (int j = 0; j < n; j++) {
                if(i == 0 || j == 0 || i == n-1 || j == n-1){
                    System.out.print("* ");
                }else{
                    System.out.print("  ");
                }
            }
            System.out.println();
        }
    }
}
```

Output:

```
* * * * * *
*         *
*         *
*         *
*         *
* * * * * *
```

Question 23: Write a program to print the following pattern:

```
     *
    ***
   *****
  *******
 *********
***********
 *********
  *******
   *****
    ***
     *
```

Code:

```
public class Pattern9{
    public static void main(String[] args) {
        int n = 6;
        for (int i = 0; i < n; i++) {
            for (int j = 1; j <= n- i -1; j++) {
                System.out.print(" ");
            }
            for (int j = 1; j <= 2 * i + 1; j++) {
                System.out.print("*");
```

```
        }
        System.out.println();
    }
    for (int i = n - 2; i >= 0; i--) {
        for (int j = 0; j < n - i - 1; j++) {
            System.out.print(" ");
        }
        for (int j = 0; j < 2 * i + 1; j++) {
            System.out.print("*");
        }
        System.out.println();
    }
  }
}
```

Output:

```
     *
    ***
   *****
  *******
 *********
***********
 *********
  *******
   *****
    ***
     *
```

Question 24: Reverse a String
Problem Statement:
Write a Java program that asks the user for a string and then prints the reverse of that string.

Code:
```
import java.util.Scanner;
public class ReverseString {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a string: ");
        String str = sc.nextLine();

        String reversed = "";
        for (int i = str.length() - 1; i >= 0; i--) {
            reversed += str.charAt(i);
        }
        System.out.println("Reversed string: " + reversed);
    }
}
```

Expected Output:
Enter a string: hello
Reversed string: olleh

Question 25: Count Vowels in a String Problem Statement:
Write a Java program that asks the user for a string and counts the number of vowels (a, e, i, o, u) in the string. The program should then print the total number of vowels.
Code:
```
import java.util.Scanner;
public class CountVowels {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a string: ");
        String str = sc.nextLine();
        int count = 0;
         str = str.toLowerCase();
        for (int i = 0; i < str.length(); i++) {
            char ch = str.charAt(i);
            if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u') {
                count++;
            }
        }
        System.out.println("The number of vowels in '" + str + "' is: " + count);
```

```
    }
}
```

Expected Output:
Enter a string: programming
The number of vowels in 'programming' is: 3

Question 26: Check if a String is a Palindrome Problem Statement:
Write a Java program that asks the user for a string and checks whether the string is a palindrome. A palindrome is a string that reads the same backward as forward (ignoring spaces and punctuation).

Code:
```
import java.util.Scanner;
public class PalindromeCheck {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a string: ");
        String str = sc.nextLine();

        String rev = "";

        for (int i = str.length()-1; i >= 0; i--) {
            rev += str.charAt(i);
        }
        if (str.equals(rev)) {
            System.out.println("The string '" + str + "' is a palindrome.");
        } else {
            System.out.println("The string '" + str + "' is not a palindrome.");
        }
    }
}
```

Expected Output:
Enter a string: madam
The string 'madam' is a palindrome.

Question 27: String Literal and Object Creation Problem Statement:
Write a Java program that creates two string variables using string literals with the same content. Then, print whether both variables point to the same object.

Code:
```
public class StringLiteral{
    public static void main(String[] args) {
        String str1 = "hello";
        String str2 = "hello";

        System.out.println("Both variables point to the same object: " + (str1 == str2));
    }
}
```

Expected Output:
Both variables point to the same object: true

Question 28: String Creation with new Keyword:
Problem Statement:
Write a Java program that creates two string objects using the new keyword with the same content. Then, print whether both objects are the same using the == operator and the .equals() method.

Code:
```
public class StringNewKeyword {
    public static void main(String[] args) {
        String str1 = new String("hello");
        String str2 = new String("hello");

        System.out.println("Using == : " + (str1 == str2));
        System.out.println("Using .equals(): " + str1.equals(str2));
    }
}
```

Expected Output:
Using == : false
Using .equals(): true

Question 29: String Concatenation and Object Creation Problem Statement:
Write a Java program that concatenates two strings using the + operator. Print whether the concatenated string is a new object or a reference to an existing string object using the == operator.

Code:

```
public class StringConcatenation  {
    public static void main(String[] args) {
        String str1 = "hello";
        String str2 = "world";
        String str3 = str1 + str2;

        System.out.println("Is str3 pointing to the same object as str1? " + (str3 == str1));
    }
}
```

Expected Output:
Is str3 pointing to the same object as str1? False

Question 30: String Pool with intern() Method Problem Statement:
Write a Java program that creates a string using the new keyword and then calls the intern() method. Print whether the interned string is pointing to the same object as the original string literal.

Code:

```
public class StringInternMethod {
    public static void main(String[] args) {
        String str1 = new String("hello");
        String str2 = str1.intern();
        String str3 = "hello";

        System.out.println("Is str2 and str3 pointing to the same object? " + (str2 == str3));
    }
}
```

Expected Output:
Is str2 and str3 pointing to the same object? true

Question 31: Multiple String Literals with Same Content:

Problem Statement:

Write a Java program that declares three string literals with the same content and prints whether all three strings refer to the same object using the == operator.

Code:

```
public class MultipleStringLiterals  {
    public static void main(String[] args) {
        String str1 = "java";
        String str2 = "java";
        String str3 = "java";
        boolean MultiString = (str1 == str2) && (str2 == str3);

        System.out.println("All strings point to the same object: " + MultiString);
    }
}
```

Expected Output:

All strings point to the same object: true